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Cloud Computing

Cloud Computing Fundamentals

Cloud Computing

Definition

On-demand delivery
Cost saving of computing
scalability
resources over flexibility
the Internet
Accessibility

characteristics:

defined by
on-demand access
Resource pooling
Elasticity
Measured service

Benifits

Provides - Scalability
results in → Cost efficiency
leads to → Global reach

Limitations

Susceptible to → Security and privacy task

Consequence of → vendor lock-in

Requires → Continuous internet connectivity

② Evolution of cloud Computing

Hardware Evolution Distributed systems

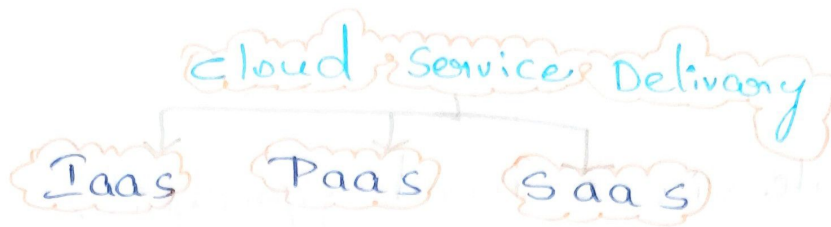
Mainframes Computing
client-server Computing
virtual Machines storage

Internet Software

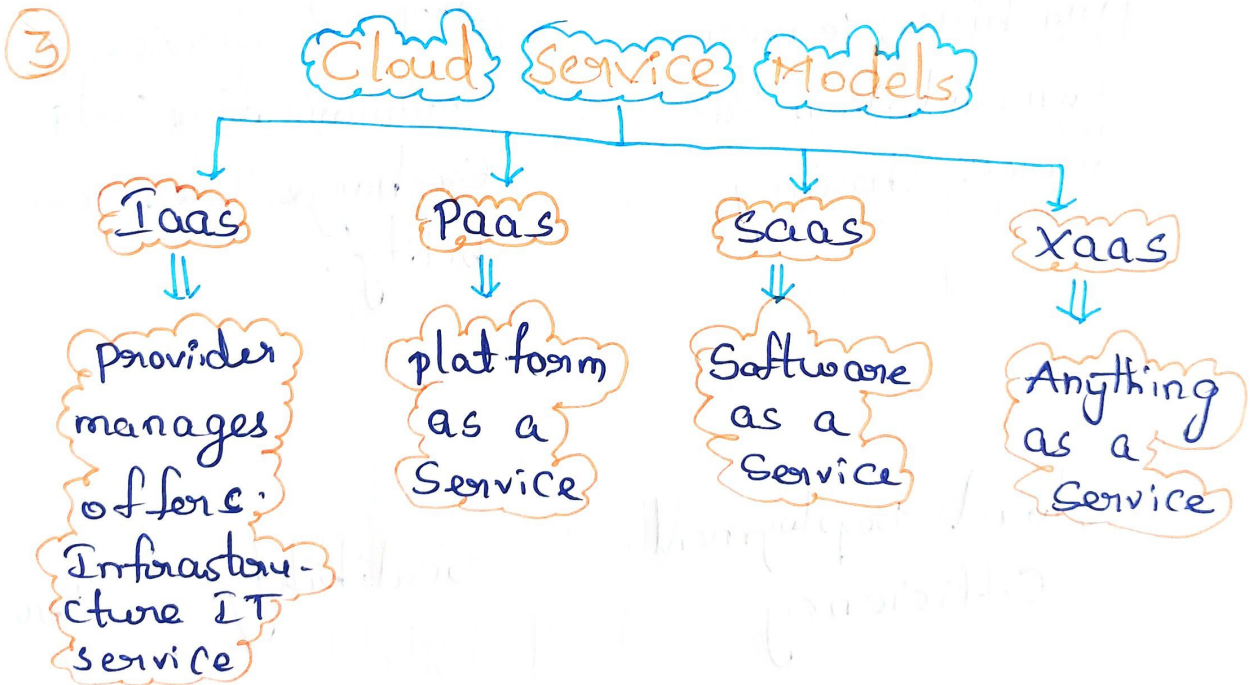
web Applications
web Services
APIs

Distributed system

Grid
cluster
Distributed



- Hardware evolution enabled → cloud computing
- * Internet software evolution supported → cloud computing.
- * Distributed systems formed the foundation of → cloud computing.
- cloud computing delivers → IaaS, PaaS, SaaS



Roles & Layers

operate via → shared responsibility

Contains → Infrastructure layer

platform layer

Application layer

users →

provider

consumer

4.

Virtualization & Web Services

Virtualization

Creates virtual machines, servers
Consolidation Resource sharing

Web Services

APIs & Service
Communication Data
Exchange Interoperability.

Enable

Cloud

Computing

Cloud Deployment
Efficiency

Scalability

Resource

* Virtualization creates → virtual Machines
web service provide → APIs and Communication

* Both enable → cloud Computing.

* cloud Computing supports → cloud deployment.

* cloud Computing achieves → Scalability and Resource Efficiency.